Many choices are possible for the conservative force field [219, 244, 342]. In case the DPD equations are used simply to control the momentum (and not as a coarse-graining device), any empirical molecular force field may be used. When DPD is used for coarse-graining, it is common to adopt a simplified pair potential, namely

$$\boldsymbol{F}_{ij}^{\mathrm{C}} = \begin{cases} \alpha\left(1 - \dfrac{r_{ij}}{r_c}\right)\hat{\boldsymbol{q}}_{ij}, & r_{ij} < r_c; \\ 0, & r_{ij} \geq r_c, \end{cases} \tag{8.82}$$

where r_{ij} is the interparticle distance, $\hat{\boldsymbol{q}}_{ij} = r_{ij}^{-1}(\boldsymbol{q}_i - \boldsymbol{q}_j)$, parameters α define the strength of the repulsion of particles i and j, and r_c is the cutoff radius. The use of a cutoff is important for efficiency reasons—in a large system, only the particles interacting at close range will generate nonzero forces of interaction so the force calculation may be substantially reduced. Both α and r_c could be allowed to be different for different particle pairs, although they are most often taken to be homogeneous. The pair potential corresponding to the force is

$$\varphi_{\mathrm{DPD}}(r) = \begin{cases} \dfrac{\alpha r_c}{2}\left(1 - \dfrac{r}{r_c}\right)^2, & r < r_c; \\ 0, & r \geq r_c. \end{cases} \tag{8.83}$$

so that the conservative potential energy is:

$$U_{\mathrm{DPD}} = \sum_{i<j} \varphi_{\mathrm{DPD}}(r_{ij}), \qquad r_{ij} = \|\boldsymbol{q}_i - \boldsymbol{q}_j\|. \tag{8.84}$$

The choice of dissipative and random forces is made so as to maintain a prescribed temperature. The typical formulas used for this purpose are

$$\boldsymbol{F}_{ij}^{\mathrm{D}} = -\gamma\omega^{\mathrm{D}}(r_{ij})(\hat{\boldsymbol{q}}_{ij} \cdot \boldsymbol{v}_{ij})\hat{\boldsymbol{q}}_{ij}, \tag{8.85}$$

$$\boldsymbol{F}_{ij}^{\mathrm{R}} = \sigma\omega^{\mathrm{R}}(r_{ij})\theta_{ij}\hat{\boldsymbol{q}}_{ij}, \tag{8.86}$$

where γ and σ parameterize the forces (again, they could be taken to be different for different pairs), and ω^{D} and ω^{R} are configuration-dependent weight functions that typically incorporate a cut-off similar to that used in the conservative force, e.g. one may take

$$\omega^{\mathrm{R}}(r) = \begin{cases} 1 - \dfrac{r}{r_c}, & r < r_c; \\ 0, & r \geq r_c, \end{cases} \tag{8.87}$$

where r_c is the same cutoff used in the conservative force. The relative velocities are denoted by $\boldsymbol{v}_{ij} = \boldsymbol{p}_i/m_i - \boldsymbol{p}_j/m_j$, and θ_{ij} is a symmetric ($\theta_{ij} = \theta_{ji}$) Gaussian